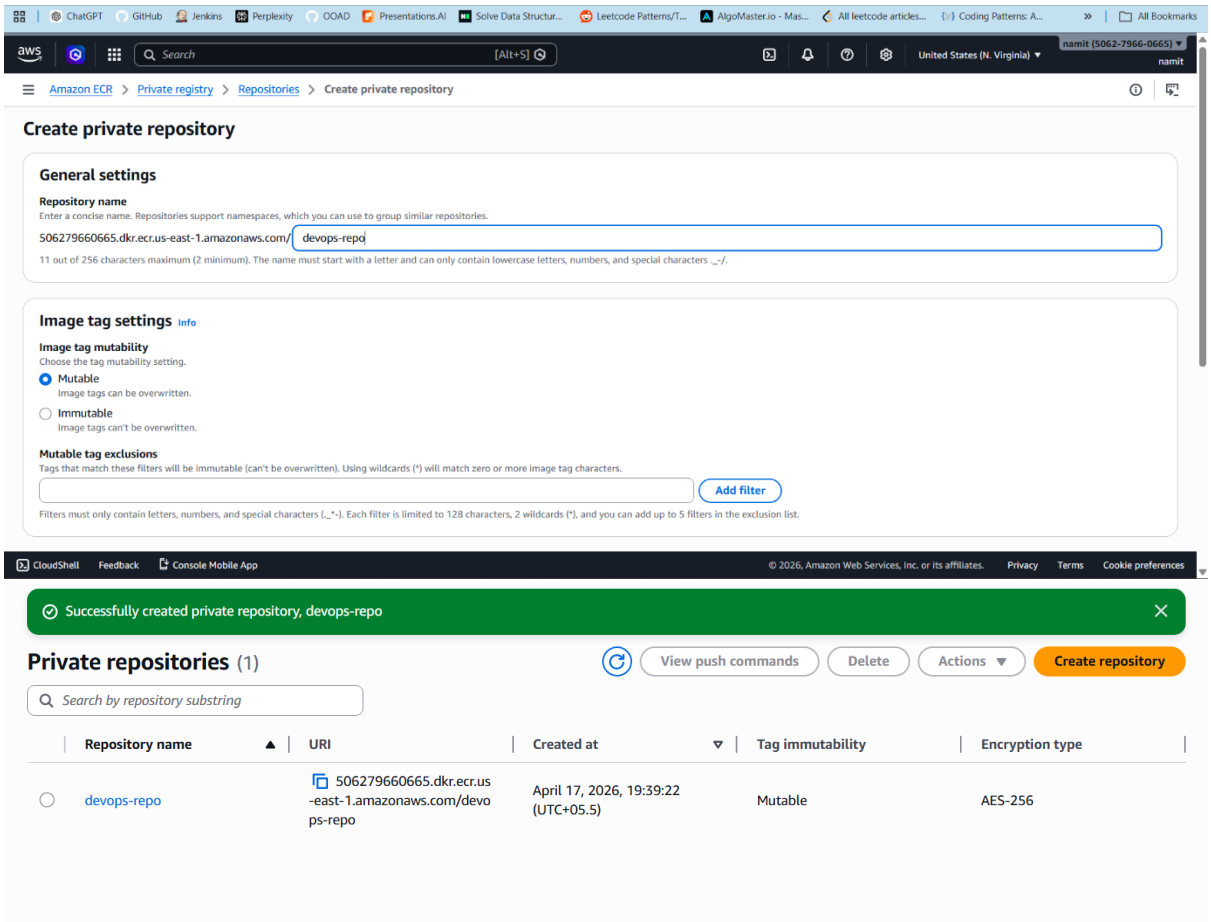
Lab Exercise 20

App Deployment Using ECS And ECR

Step 1: Create ECR Repository

- Go to AWS Console
- Search Elastic Container Registry (ECR)
- Click Create Repository
- Give name (example: my-web-app)
- Click Create



Step 2: Configure aws

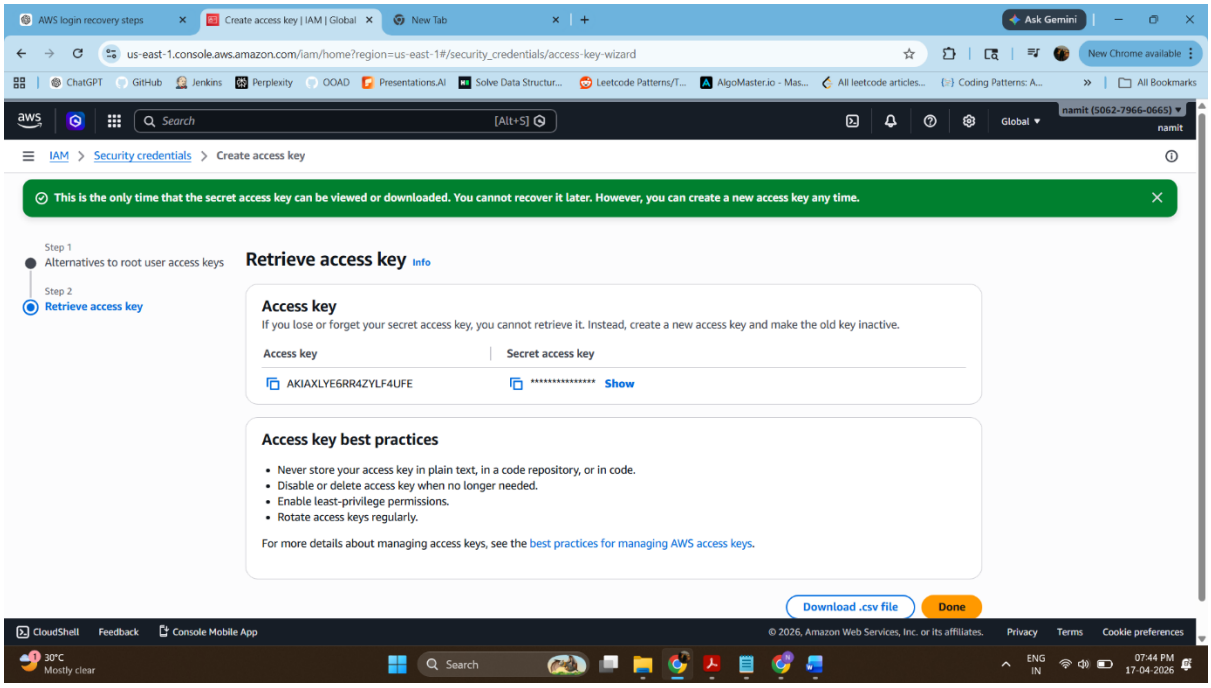
Create access key

```
Command Prompt

Microsoft Windows [Version 10.0.26200.8117]
(c) Microsoft Corporation. All rights reserved.

C:\Users\namit>aws configure
AWS Access Key ID [****************PMRL]: AKIAXLYE6RR4ZYLF4UFE
AWS Secret Access Key [****************TGmY]: 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
Default region name [ap-south-1]:
Default output format [json]:

C:\Users\namit>
```



Step 3: Authenticate Docker to ECR

```
PS C:\Users\namit> aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 506279660665.dkr.ecr.us-east-1.amazonaws.com

Login Succeeded
PS C:\Users\namit>
```

Step 4: Build Docker Image

Run this :

```
PS C:\Users\namit\Downloads\my-web-app\my-web-app> docker build -t devops-repo .
[+] Building 27.9s (8/8) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.1s
 => => transferring dockerfile: 102B                                                                                0.0s
 => [internal] load metadata for docker.io/library/httpd:latest                                                     4.6s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [1/3] FROM docker.io/library/httpd:latest@sha256:bdba5c86022f2d6ad0783168b07fe80541fe45e97b434b13ba8c959a505  19.8s
 => => resolve docker.io/library/httpd:latest@sha256:bdba5c86022f2d6ad0783168b07fe80541fe45e97b434b13ba8c959a5050  0.1s
 => => sha256:4e51448030490b8604d7662b05848e89e3563fe3b955159f7ec54d4d48a4893f 289B / 289B                         0.8s
 => => sha256:61e1798e23ba12f14505e225391338db290ee6cecfb6321c6c6b92ac8b546dcb 13.45MB / 13.45MB                   7.7s
 => => sha256:a838a4be55ced5f2d7208c56a3a13ecbd15110574ee338337c50737aef163fdf 2.00MB / 2.00MB                     2.3s
 => => sha256:8261ede62f8c1c85c3e4b2ae3a9ae5a586b84fcc61b60c110ee65c9c523b49c7 146B / 146B                         1.5s
 => => sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776fab9a605336f5067d6e9ecb5976 29.78MB / 29.78MB                   8.8s
 => => extracting sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776fab9a605336f5067d6e9ecb5976                           6.4s
 => => extracting sha256:8261ede62f8c1c85c3e4b2ae3a9ae5a586b84fcc61b60c110ee65c9c523b49c7                           0.2s
 => => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1                           0.1s
 => => extracting sha256:a838a4be55ced5f2d7208c56a3a13ecbd15110574ee338337c50737aef163fdf                           1.2s
 => => extracting sha256:61e1798e23ba12f14505e225391338db290ee6cecfb6321c6c6b92ac8b546dcb                           2.0s
 => => extracting sha256:4e51448030490b8604d7662b05848e89e3563fe3b955159f7ec54d4d48a4893f                           0.1s
 => [internal] load build context                                                                                   0.1s
 => => transferring context: 180B                                                                                   0.0s
 => [2/3] WORKDIR /usr/local/apache2/htdocs/                                                                        1.1s
 => [3/3] COPY index.html .                                                                                         0.5s
 => exporting to image                                                                                              1.4s
 => => exporting layers                                                                                             0.7s
 => => exporting manifest sha256:ed0d10492343b3c6605bdefaea81cbf6ef87ed89c8921fdc0290160b7a3521e0                   0.1s
 => => exporting config sha256:94fdfba2978285be5bdadd41642d7ff76818e904100359e59a277524a1d29b96                     0.1s
 => => exporting attestation manifest sha256:f908edc9ff4ba6db06762b1f50db2612667873f7f9924d91fc497e4db443fbcc       0.1s
 => => exporting manifest list sha256:130af5295f080e6688735335f3b20db202df9799ed230423d43e2fac3fbdaedc              0.0s
 => => naming to docker.io/library/devops-repo:latest                                                               0.0s
 => => unpacking to docker.io/library/devops-repo:latest                                                            0.2s
PS C:\Users\namit\Downloads\my-web-app\my-web-app>
```

Step 5: Tag Docker Image

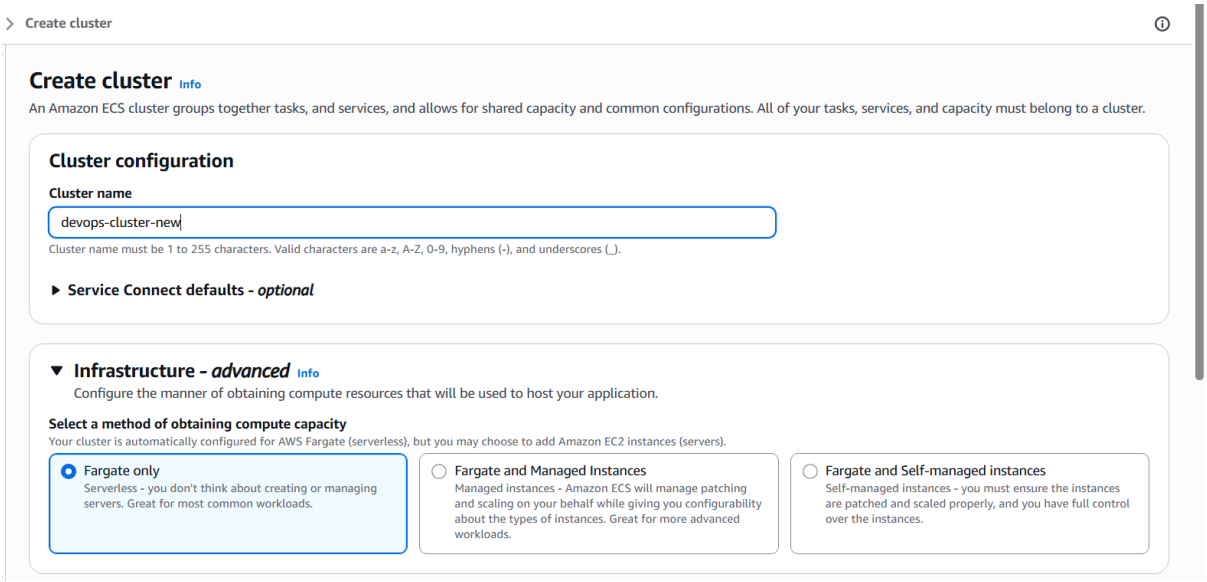
```
PS C:\Users\namit\Downloads\my-web-app\my-web-app> docker tag devops-repo:latest 506279660665.dkr.ecr.us-east-1.amazonaws.com/devops-repo:latest
PS C:\Users\namit\Downloads\my-web-app\my-web-app>
```

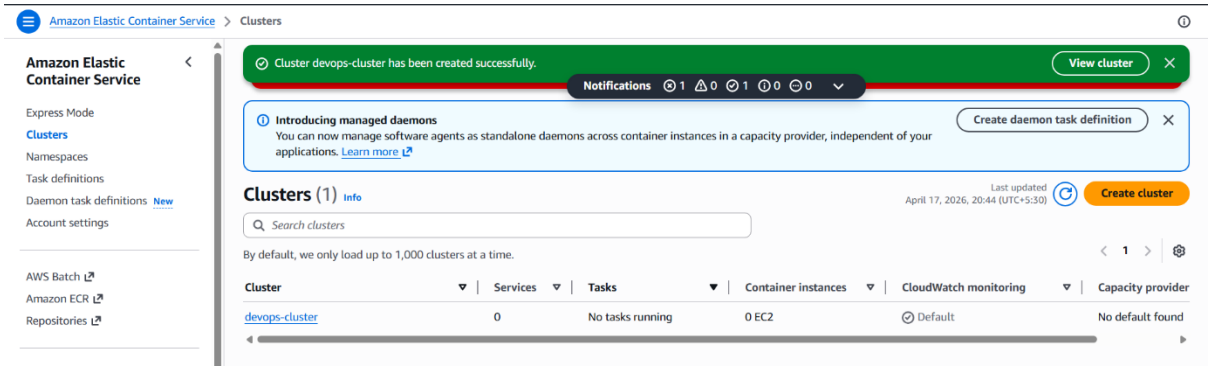
Step 6 : Push Docker Image

```
PS C:\Users\namit\Downloads\my-web-app\my-web-app> docker push 506279660665.dkr.ecr.us-east-1.amazonaws.com/devops-repo:latest
The push refers to repository [506279660665.dkr.ecr.us-east-1.amazonaws.com/devops-repo]
c8ed30fa4d32: Pushed
4f4fb700ef54: Pushed
5435b2dcdf5c: Pushed
4e5144803049: Pushed
61e1798e23ba: Pushed
8261ede62f8c: Pushed
a838a4be55ce: Pushed
5c013e329e54: Pushed
latest: digest: sha256:130af5295f080e6688735335f3b20db202df9799ed230423d43e2fac3fbdaedc size: 856
PS C:\Users\namit\Downloads\my-web-app\my-web-app>
```

Step 7: Create Cluster

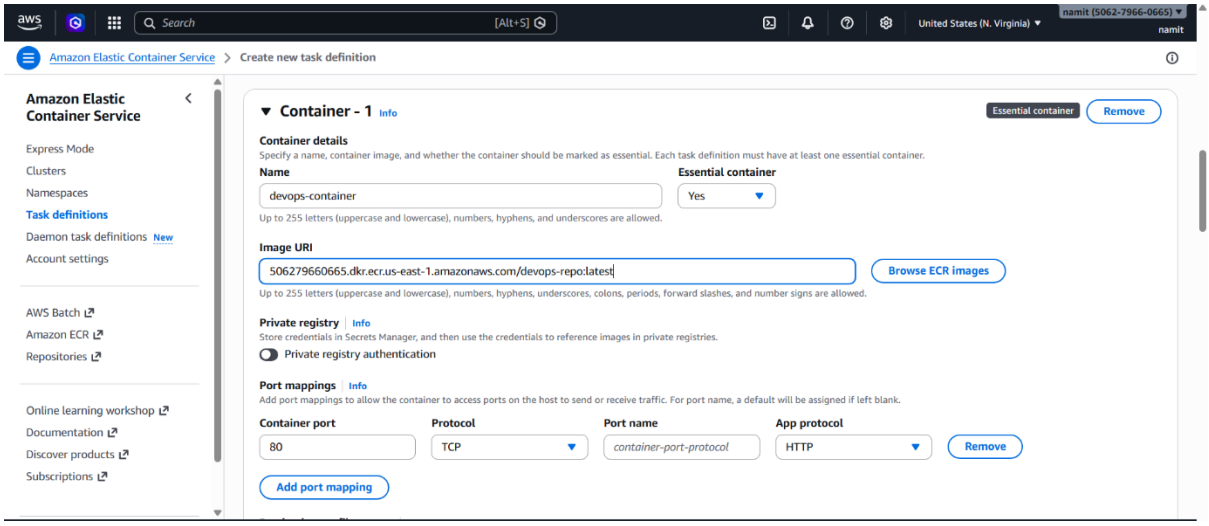
1. Go to ECS
2. Click Create Cluster
3. Select Fargate (Serverless)
4. Cluster Name: devops-cluster-new
5. Click Create

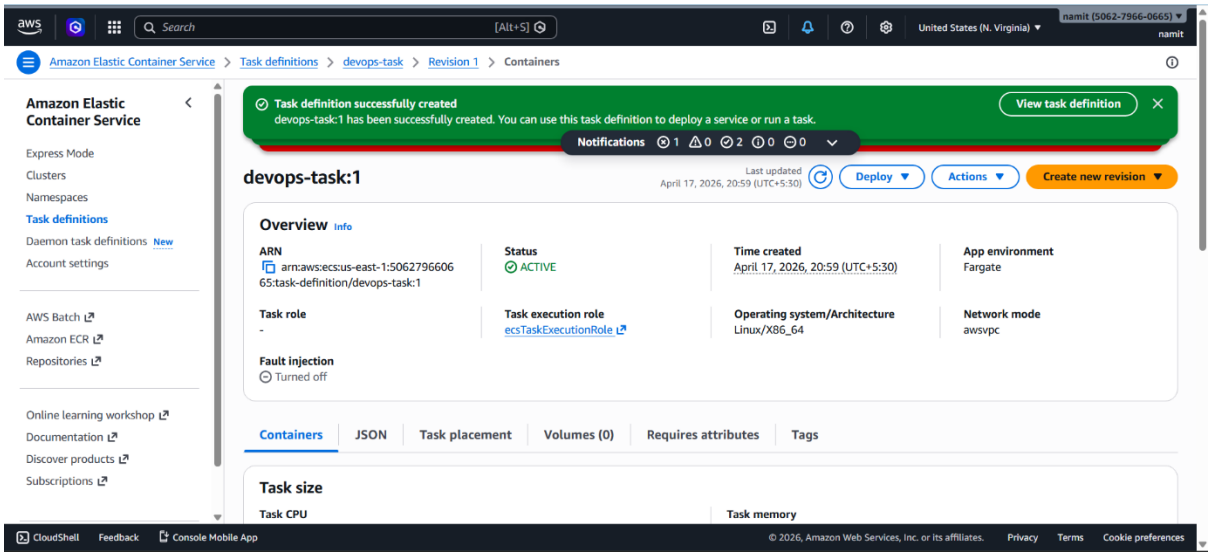




Step 8: Create Task Definition

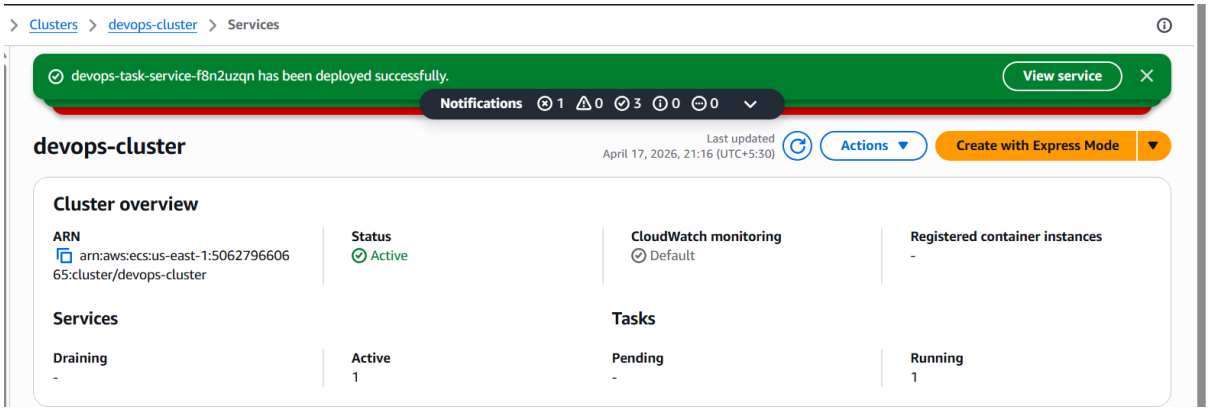
1. Launch Type → Fargate
2. Container Image → Paste ECR URI



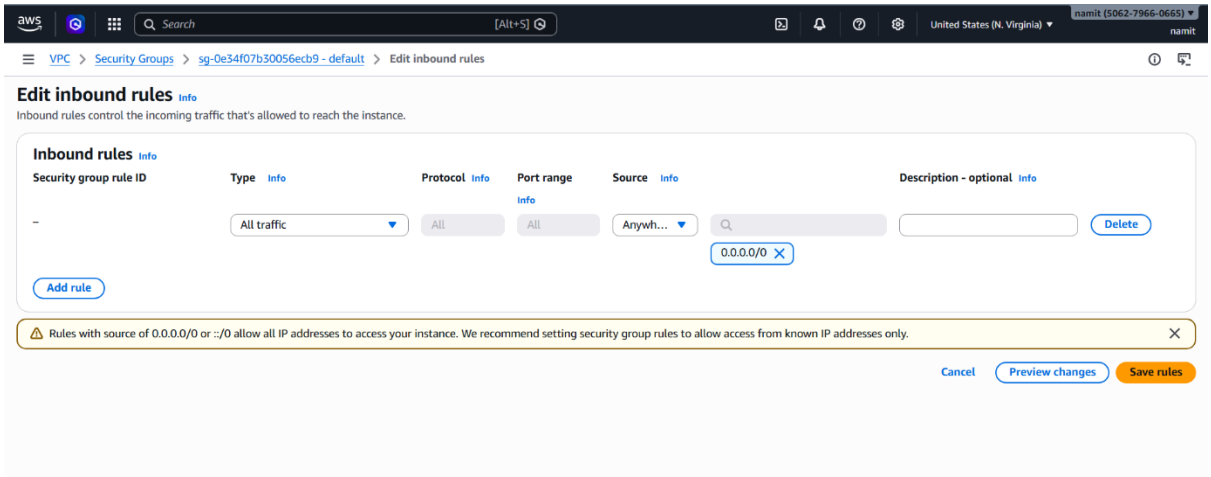


Step 9: Create Service

1. Go to Cluster
2. Click Create Service



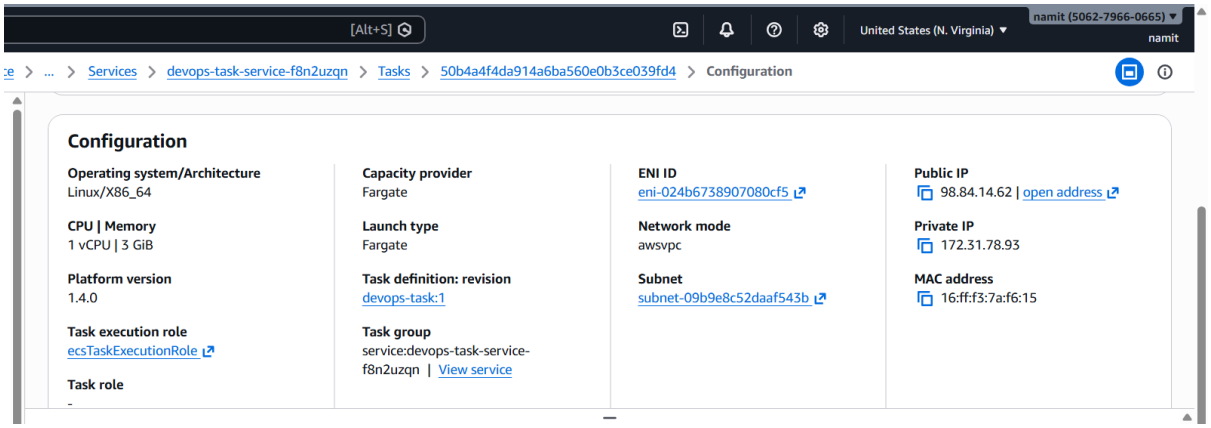
Step 10: Turn on all traffic in security broke



Step 11 : Verify Deployment

1. Go to Load Balancer
2. Copy DNS Name
3. Paste in Browser

http://98.84.14.62/


[Alt+S]
United States (N. Virginia)
namit (5062-7966-0665)
namit
Services > devops-task-service-f8n2uzqn > Tasks > 50b4a4f4da914a6ba560e0b3ce039fd4 > Configuration
Configuration
Operating system/Architecture
Linux/X86_64
CPU | Memory
1 vCPU | 3 GiB
Platform version
1.4.0
Task execution role
ecsTaskExecutionRole
Task role
Capacity provider
Fargate
Launch type
Fargate
Task definition: revision
devops-task:1
Task group
service:devops-task-service-f8n2uzqn | View service
ENI ID
eni-024b6738907080cf5
Network mode
awsvpc
Subnet
subnet-09b9e8c52daaf543b
Public IP
98.84.14.62 | open address
Private IP
172.31.78.93
MAC address
16:ff:f3:7a:f6:15

Not secure
98.84.14.62
New Chrome available
ChatGPT
GitHub
Jenkins
Perplexity
OOAD
Presentations.AI
Solve Data Structur...
Leetcode Patterns/T...
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